

M. Thufail Alwannabil Samas

Applied AI & Automation | Data Scientist

Surabaya, East Java, Indonesia | +62 877-6313-3209 | mthufailalwannabils@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Mathematics graduate and BNSP-certified Data Scientist focused on applied AI, workflow automation, and integration development. Built an evaluated invoice approval system using n8n, Gemini API, FastAPI, PostgreSQL, authenticated webhooks, deterministic validation, human approval, and an idempotent ERP sandbox. Experienced in building Python/PostgreSQL operational pipelines, REST APIs, dashboards, and technical learning materials.

CORE SKILLS

AI Automation & Integration: n8n, REST APIs, webhooks, JSON, authentication, idempotency, ERP integration patterns

Applied AI: Gemini API, LLM structured output, document extraction, prompt design, evaluation, human-in-the-loop controls

Engineering: Python, SQL, PostgreSQL, FastAPI, Pydantic, unittest, Git, HTML, CSS, JavaScript

Data & Machine Learning: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow/Keras, data validation, time-series analysis, model evaluation

SELECTED AI AUTOMATION PROJECT

AI Invoice Approval Automation

Personal Portfolio Project

Applied AI Automation Developer

July 2026

- Designed and built an end-to-end synthetic vendor-invoice workflow spanning authenticated n8n intake, Gemini document extraction, deterministic Python validation, finance review, amount-based human approval, and ERP sandbox handoff.
- Implemented FastAPI services and PostgreSQL workflow state with duplicate prevention, database-enforced transitions, original extraction and correction history, append-only audit events, role-specific access tokens, and idempotent ERP retries.
- Evaluated Gemini on a locked 20-PDF synthetic corpus, measuring 359/360 field matches (99.72%), 19/20 exact documents, and 19/20 correct validation routes; preserved the failed baseline and diagnosed an ambiguous numeric date that changed routing.
- Added a source-date ambiguity guard, verified the unchanged failure case through a separate live recheck, and maintained 53 passing Python/PostgreSQL tests plus 35 n8n production-webhook checks with architecture, runbook, user, and evaluation documentation.

PROFESSIONAL EXPERIENCE

PT Aplikasi Lintasarta

Surabaya, Indonesia

Junior Programmer

January 2026 - April 2026

- Developed a project-stage DWDM monitoring and assurance system for approximately 400 directional sensors, integrating 5-minute PRTG telemetry with ENIMS network metadata through an incremental Python/PostgreSQL pipeline and API-served operational views.
- Implemented data-quality, missing-data, failure-event, change-point, gradual-degradation, paired-sensor, and priority logic while keeping invalid observations outside statistical analysis.
- Built a Python-backed HTML/CSS/JavaScript dashboard with Operations Board, Event Explorer, and Management Overview views for investigation and monitoring handoff.

SMP GIKI 3 Surabaya*Coding and AI Teacher*

Surabaya, Indonesia

August 2025 - December 2025

- Taught Python fundamentals and introductory AI concepts to Grade 8 and 9 students through practical coding and image-classification exercises.
- Designed lesson materials, quizzes, demonstrations, and plain-language technical explanations to support learners with different levels of programming experience.

Stasiun Meteorologi Maritim Kelas II Tanjung Perak (BMKG)*Data Science Intern*

Surabaya, Indonesia

February 2024 - June 2024

- Developed an internal one-page extreme-weather dashboard using HTML, CSS, JavaScript, PHP, MySQL, and API-based periodic updates to consolidate warnings, tables, charts, and filters.
- Processed cloud-prediction and Himawari IR imagery and built a 3-layer BiLSTM rainfall model using 7-day sequences and an 80:20 chronological split, achieving MAAPE 0.8073 and RMSE 10.2734 mm.

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)*Programming Teaching Assistant*

Surabaya, Indonesia

August 2022 - July 2024

- Guided undergraduate students for four semesters in MATLAB, Python, debugging, data handling, algorithms, and introductory machine learning while providing structured technical feedback.

EDUCATION**Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)***Bachelor of Mathematics (S.Mat.), GPA 3.46/4.00*

Surabaya, Indonesia

August 2021 - January 2025

Thesis: Rainfall Forecasting in Sumenep Regency Using XGBoost and Grid Search. Completed the four-year program in seven semesters.

ADDITIONAL DATA & AI PROJECTS

Rainfall Forecasting with XGBoost and Grid Search: Built a next-day forecasting workflow with fold-wise scaling, 10-fold TimeSeriesSplit, and 243 hyperparameter configurations, achieving MAAPE 0.9152 and RMSE 11.9566 mm.

Thyroid Cancer Recurrence Classification: Built a leakage-controlled XGBoost workflow on 383 UCI records with fold-wise Information Gain and 10-fold cross-validation, achieving 97.25% accuracy and 98.60% ROC-AUC.

PUBLICATION & CERTIFICATION

First-author peer-reviewed article: "Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode Extreme Gradient Boosting (XGBoost) dan Algoritma Grid Search," MIND Journal, Vol. 11, No. 1, pp. 30-43, June 2026. [DOI: 10.26760/mindjournal.v11i1.30-43](https://doi.org/10.26760/mindjournal.v11i1.30-43)

Data Scientist - Badan Nasional Sertifikasi Profesi (BNSP), valid September 2024 to September 2027.